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Reliability screening for vehicle routing with time windows: a solver-independent alternative to travel-time padding

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Abstract

Routing plans built on nominal travel times fail under congestion and random delay, and the usual safeguard of padding every trip wastes capacity on safe routes. This paper adds a statistical reliability check to any solver for the vehicle routing problem with time windows. The solver generates candidate plans with internal buffers that leave customer windows unchanged. A plan is admitted only if a 95% Wilson lower bound on the probability that at least 95% of customers are served on time reaches 0.95; admitted plans are validated on fresh scenarios. Capping each buffer at the customer's direct-reachability limit removes a failure that also blocks budgeted-robust models. On synthetic instances with 20–200 customers, screening beat 20% padding by 8.3–21.4 percentage points (not significant at 200 customers). In three timestamped protocols, seven of eight primary hypotheses held; a conservative-speed fallback did not measurably help. Results depend on the assumed delay model.

Keywords: vehicle routing; time windows; stochastic travel times; time-dependent travel times; chance constraints; simulation; reproducibility

1. Introduction

The vehicle routing problem with time windows (VRPTW) asks a fleet to serve customers within agreed windows, subject to vehicle capacity and shift length (Solomon, 1987; Toth and Vigo, 2014). Most planning tools solve it on one fixed matrix of travel times. Real travel times change with the time of day and with random delays, and a delay early in a route pushes every later customer towards the end of its window. A plan that is optimal on nominal times takes no account of this.

The simplest safeguard is to plan at a slower speed than expected. Padding every trip by a fixed percentage is easy to explain and needs no new software, but it protects off-peak trips as much as peak

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ones and adds distance and vehicles whether or not a route is at risk. Stochastic and robust VRPTW models avoid that waste by building the uncertainty into the optimization (Errico et al., 2018; Agra et al., 2013; Gendreau et al., 2016). They need a solver written for their uncertainty model, and most operators run a general solver that they cannot easily replace.

This paper studies a procedure that sits between the two. A standard VRPTW solver proposes a family of candidate plans, each planned against internal deadlines contracted by a buffer, while the windows promised to customers stay fixed. Each candidate is simulated on a bank of scenarios drawn from a time-dependent, correlated delay model, and it is admitted only if a one-sided Wilson lower bound on the probability of a good service day reaches a target. Among admitted plans the procedure returns the best under the planner's own objective, and it reports that plan's reliability from an independent validation bank that played no part in the choice. The solver and its objective are left as they are; reliability is judged after the solve, by the same rule for every solver.

The paper makes three contributions.

1. It specifies this screening-and-validation procedure and identifies a failure that internal buffers create: a contracted deadline can make a customer unreachable even by a direct trip, so the solver finds no plan. Capping each customer's buffer at its direct-reachability limit removes that failure. The same obstruction appears in a budgeted-robust VRPTW model, and the analogous relief removes it there too.
2. It gives matched evidence on when screening beats uniform padding and at what cost. In a development study on 53 synthetic instances with 20 to 200 customers, the screened procedure was 8.3–21.4 percentage points more reliable than 20% padding, depending on size; at 200 customers the interval included zero. Three protocols, each publicly timestamped with frozen code before its runs (Ali, 2026a,b,c), then tested the main findings on new instances, with OR-Tools (Perron and Furnon, 2026) and hybrid genetic search (HGS; Vidal, 2022; Wouda et al., 2024) as generators. Seven of their eight primary hypotheses met their decision criteria, and the paper reports the one that did not.
3. It places the procedure against a budgeted-robust optimizer (Bertsimas and Sim, 2004; Agra et al., 2013) and tests a fallback rule for instances where no plan passes. The robust plans with reachability relief, screened in the same way, were more reliable on instances with up to 100 customers but used more vehicles; restricted to the comparator's fleet, their advantage was no longer measurable. Returning the best-screened candidate when nothing is admitted raised pooled service in a prespecified test on 53 further instances.

The instances are synthetic and the delay model is assumed, so the evidence shows repeatability within one setting. Every number in the paper is regenerated from the archived records, and the full model, every result table and further checks are in the Supporting Information (SI).

Section 2 reviews related work. Section 3 defines the planning model, the evaluation model and the procedure. Section 4 describes the experiments and Section 5 the results. Section 6 discusses implications and limitations, and Section 7 concludes.

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2. Related work

Time-dependent routing needs care because a speed multiplier applied at departure can break first-in-first-out (FIFO). Ichoua et al. (2003) introduced the integrated speed-profile construction used here, and Ghiani and Guerriero (2014) analysed its properties. Stochastic VRPTW research treats uncertain travel and service times through chance constraints, recourse, approximate feasibility tests and data-driven formulations (Laporte et al., 1992; Kenyon and Morton, 2003; Li et al., 2010; Errico et al., 2018; Bomboi et al., 2022; Serrano et al., 2026). Ehmke et al. (2015) embed per-customer service levels in a routing method; Taş et al. (2014) jointly model time dependence, stochastic travel and soft windows; and Bomboi et al. (2022) check route feasibility under correlated, time-dependent travel times. Jabali et al. (2015) introduced self-imposed time windows, and Vareias et al. (2019) assign windows under stochastic travel times with per-customer chance constraints. These are the closest analogues of the internal contraction used here, but both build the buffers into a dedicated optimization and judge reliability customer by customer. Neither can be added to an unmodified solver, and neither bounds the probability of a whole service day.

A second line builds the uncertainty into the optimization. Agra et al. (2013) solve a robust VRPTW over a budgeted uncertainty set (Bertsimas and Sim, 2004); Jaillet et al. (2016) minimize a requirements violation index; and Zhang et al. (2019) develop a riskiness index with an exact algorithm. Surveys by Gendreau et al. (1996) and Gendreau et al. (2016) cover stochastic routing more broadly. This paper compares the procedure with padding and with one budgeted robust model; it does not compare it with a chance-constrained or distributionally robust VRPTW solver.

This paper makes its statistical claim after the solve. It does not build a feasibility approximation into a specialist search. It takes routes from a replaceable nominal solver, searches a family of internal buffers on one scenario bank and checks the chosen plan once on a held-out bank. Its service event limits the share of late customers across the whole day, which is closer to the plan-level risk measures of Kenyon and Morton (2003) than to per-customer constraints. Windows offered to customers never change. Monte Carlo approximations of chance constraints need a separate check of the chosen solution (Ahmed and Shapiro, 2008), which the validation bank provides, and the Wilson bound is used because normal-approximation intervals behave poorly near one (Wilson, 1927; Brown et al., 2001).

3. Model and procedure

3.1. Planning model

Let $N = \{0, 1, \ldots, n\}$ be the nodes, with depot 0 and customers C, and let V be a homogeneous fleet with capacity Q and maximum shift duration $H^{\max}$. Customer i has demand δ_i, service time s_i and offered window $[l_i, u_i]$ for the start of service. Planning uses nominal distances d_{ij} and times $\bar{t}_{ij}$. A plan assigns customers to routes, fixes the visit order and sets a depot departure d_v and return r_v for each vehicle v, with service start a_i at each customer. The full formulation is in Section S1; there $x_{ijv} = 1$ if

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vehicle v drives arc (i, j) and $y_v = 1$ if vehicle v is used. Four criteria describe a plan:

$$F_1 = c_f \sum_v \sum_{i,j} d_{ij} x_{ijv}, \quad F_2 = \sum_{i \in C} \max\{0, s^* - (u_i - a_i)\}, \quad F_3 = \sum_v (r_v - d_v), \quad F_4 = \sum_v y_v, \tag{1}$$

namely distance cost at c_f per kilometre, slack deficit against a target slack s^*, driver duration and fleet use. Each criterion is scaled by the range $[L_k, U_k]$ it takes over four single-criterion anchor solutions, and the candidate objective is

$$Z = \sum_{k=1}^{4} w_k \, \frac{F_k - L_k}{\max\{U_k - L_k, 1\}}. \tag{2}$$

The weights $\mathbf{w} = (0.5693, 0.2643, 0.1055, 0.0609)$ are a fixed experimental vector chosen before any experiment; their origin and a sensitivity analysis are in Section S12 and Section 5.6. The objective shapes the candidates and ranks the admitted ones, but the admission test does not use it, and any scalar objective could replace it.

3.2. Evaluation model

The day is split into periods $[T_k, T_{k+1})$ with speeds $\nu_k > 0$. The deterministic travel time from i to j when leaving at time τ is the smallest $\Delta \geq 0$ with

$$\int_{\tau}^{\tau + \Delta} \nu(u) \, du = d_{ij}, \tag{3}$$

which preserves FIFO (Ichoua et al., 2003). In scenario ω the realised travel time is $t_{ij}(\tau) \exp\{\varepsilon_{ij}(\omega)\}$, with

$$\varepsilon_{ij}(\omega) = \sigma\big(\sqrt{\rho}\, \xi_0(\omega) + \sqrt{1 - \rho}\, \xi_{ij}(\omega)\big), \tag{4}$$

where ξ_0 is a common standard normal factor drawn once per scenario and ξ_{ij} is an arc-specific one. The multiplier has median one, so the deterministic profile gives the median travel time. Each arc draws from its own stream keyed by the bank seed and its endpoints, so plans evaluated on one bank see the same disturbances on shared arcs. A fixed set of simulated days drawn in this way is called a scenario *bank*. Every vehicle leaves at its planned time in every scenario.

3.3. Screening and validation

For plan R in scenario ω, let $q(R, \omega)$ be the share of customers whose service starts by u_i. The service event is $J(R, \omega) = \mathbf{1}\{q(R, \omega) \geq \kappa\}$ with $\kappa = 0.95$, and the target is $\Pr[J = 1] \geq \alpha$ with $\alpha = 0.95$. Requiring every customer to be on time would be a poor target for large instances, because that probability

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falls with the number of customers even for good plans; it is reported but not used for admission.

For each buffer β in a grid, the solver plans against the contracted deadline $\max\{l_i, u_i - \beta\}$ and minimizes (2), with the anchor ranges fixed at their uncontracted values. Each candidate is simulated on a screening bank of N_s scenarios, and with empirical success $\hat{p}$ its one-sided Wilson lower bound at level $1 - \eta$ is

$$L(\hat{p}) = \frac{\hat{p} + z^2/(2N_s) - z\sqrt{\hat{p}(1-\hat{p})/N_s + z^2/(4N_s^2)}}{1 + z^2/N_s}, \qquad z = z_{1-\eta}. \tag{5}$$

A candidate is admitted when $L(\hat{p}) \geq \alpha$, and the admitted candidate with the smallest Z is selected. It is then simulated once on a larger, untouched validation bank, which only reports its reliability. Scoring always uses the offered windows u_i. If no candidate is admitted, a fallback rule fixed in advance returns a plan, so the procedure has an output on every instance. The results therefore distinguish the validated reliability of admitted plans from the procedure-level reliability over all instances, fallbacks included; the latter, averaged over instances, is called *pooled service*. The bound holds for each candidate separately; a family-wise version over the grid is examined in Section 5.6. The procedure is:

1. For each buffer $\beta \in \{0, 5, \ldots, 60\}$ minutes, solve the planning model with deadlines $\max\{l_i, u_i - \beta\}$ (or (6) when capped).
2. Simulate each candidate on the screening bank and compute $L(\hat{p})$.
3. If some candidate has $L(\hat{p}) \geq \alpha$, select the admitted candidate with the smallest Z; otherwise apply the fallback rule.
4. Simulate the selected plan once on the validation bank and report its service probability.

3.4. Reachability-capped contraction

Contraction can make a customer unreachable even by a direct trip from the depot, in which case no plan exists at that buffer. Let $e_i = \max\{l_i, \bar{t}_{0i}\}$ be the earliest direct service time. The capped procedure plans against

$$u_i^{\mathrm{cap}}(\beta) = \min\big\{u_i, \max\{l_i, u_i - \beta, e_i\}\big\}, \tag{6}$$

so contraction alone never removes a customer that was reachable before. The cap does not guarantee that a joint routing exists or that the target is met. A shortest-path relaxation certifies when a contracted model is infeasible, which separates proven infeasibility from search failure (Section S4.1).

3.5. Comparators

Conservative speed plans for distance at 33.3 km/h instead of 40 km/h, which pads every trip by 20%. *Screened conservative speed* applies the same screen to conservative-speed candidates. The *original procedure* screens contracted candidates without the cap, and the *capped procedure* uses (6). The candidate generator is OR-Tools (Perron and Furnon, 2026) or the HGS of PyVRP (Wouda et al., 2024), which solves the same planning model but cannot express the slack-deficit term during generation.

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Table 1
Instances and parameters of the development study. The confirmatory studies use the same sizes with the seeds shifted (Section 4).

Parameter	Value
Customers (vehicles available; instances)	20 (6; 20), 50 (14; 15), 100 (26; 10), 200 (40; 8)
Nominal speed; shift	40 km/h; 600 min from 06:00
Speed divisors in the shift	0.9 (06:00–07:00), 1.2 (07:00–09:00), 1.0 (09:00–16:00)
Capacity; service time; target slack s^*	1000; 10 min; 15 min
Disturbance σ; latent correlation ρ	0.20; 0.50 (sweep $\sigma \in \{0.1, 0.2, 0.3\}$, $\rho \in \{0, 0.5, 0.9\}$)
Service fraction κ; target α; Wilson η	0.95; 0.95; 0.05 (one-sided)
Screening; validation; sweep scenarios	1000; 5000; 1000 per cell
Buffer grid	$\beta \in \{0, 5, \ldots, 60\}$ min
Solver budgets	OR-Tools 5 s per candidate; HGS 10,000 iterations

The robust comparator is a VRPTW with budgeted uncertainty (Bertsimas and Sim, 2004; Agra et al., 2013): on each arc the travel time may rise from $\bar{t}_{ij}$ to $\bar{t}_{ij} + \lceil \theta \bar{t}_{ij} \rceil$, and a route must stay feasible whenever at most Γ of its arcs do so; *box uncertainty* is the case in which every arc may take its upper value. Feasibility is checked exactly with the dynamic programme of Agra et al. (2013) inside a large neighbourhood search written for this study, which minimizes distance and stops after a fixed number of iterations, so it is deterministic. Robust models meet the same obstruction as contraction: a customer that cannot be reached on time by a direct trip at its upper travel time makes every plan infeasible. The *relieved* variant raises such a customer's robust deadline to its worst-case direct start while every original window must still hold under nominal times. *Screened robust* passes all robust plans of an instance through the same screen and returns the shortest admitted one, or the one with the highest screening success. Section S8 gives the details.

Three fallback rules are compared: the uncontracted slack-aware plan (*slack-aware*), the generator's conservative-speed plan (*conservative*), and the candidate of the procedure's own family with the highest screening success (*best-screened*). None of them uses a validation outcome. Unless stated otherwise, the OR-Tools procedures use the slack-aware fallback and the HGS procedures the conservative one, as fixed in their protocols.

4. Experimental design

Instances place a central depot and uniformly distributed customers on a 100×100 grid. Demand is uniform on 10–49 units, service takes 10 minutes, window widths are uniform on 90–149 minutes, capacity is 1000 and the shift lasts 600 minutes from 06:00. Table 1 lists the instance sizes and the model constants. The planned shift has a morning peak from 07:00 to 09:00, and the speed profile continues into the evening so that overrunning routes are still evaluated.

The development study ran every configuration on 53 instances with fixed seeds, and its findings are exploratory. Within an instance all configurations share the same screening and validation banks, so every comparison is paired. Across-instance intervals use a paired bootstrap with 20,000 resamples and a fixed seed. A configuration without a plan scores zero, so a failure counts against the configuration that failed. Software errors are recorded and block the analysis; they are never scored as routing outcomes.

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Table 2
The studies. Seeds of the confirmatory instances are the development seeds shifted by a fixed offset per study.

Study	Instances	Generators	Question
Development	53	OR-Tools	Screening versus padding; capping; robust comparator (exploratory)
Protocol 1 (H1–H4)	53 new	OR-Tools	Do screening and capping beat padding on new instances?
Protocol 2 (G1–G4)	53 new	HGS, OR-Tools	Does the gain survive a stronger generator? Does a conservative fallback help?
Protocol 3A (R1–R4)	90 earlier	Robust, HGS, OR-Tools	Are screened robust plans more reliable, and at what fleet cost?
Protocol 3B (F1–F4)	53 new	OR-Tools, HGS	Does the best-screened fallback raise service?

Three confirmatory protocols followed. Each fixed its hypotheses, tests, decision rules, failure policy and instance seeds, and was committed to a public repository with a SHA-256 manifest of the code that runs and analyses it, and archived on Zenodo, before any of its instances was solved (Ali, 2026a,b,c). The analysis programs stop unless every record is present and every frozen file matches its manifest. Protocol 1 tested the OR-Tools procedures on 53 new instances. Protocol 2 tested HGS as the generator and the conservative fallback on 53 further instances. Protocol 3 had two parts: part A repeated the robust comparison on the 90 confirmatory instances with up to 100 customers, and part B tested the best-screened fallback on 53 new instances. Table 2 summarizes the studies. In each study the two primary hypotheses use 97.5% intervals (Bonferroni) and are confirmed only if the interval excludes zero in the stated direction; secondary hypotheses use 95% intervals and cannot rescue a primary one. All instances come from the same generator and delay model, so the protocols test repeatability in that setting rather than external validity. Hardware, software versions and the course of each run are in Sections S6, S7 and S9.

5. Results

5.1. Screening against padding

Table 3 gives the paired contrasts with conservative speed on the 46 development instances where conservative speed found a plan. The complete original procedure was 8.3–21.4 points more reliable, depending on size, although the interval at 200 customers includes zero. The slack-aware objective without screening was less reliable than padding at every size, so the gain comes from the candidate family and the screen acting together. Screening conservative-speed candidates also helped, which makes the fixed 20% pad the weaker of the two padding baselines.

The advantage grows with noise when delays are independent (Fig. 1). With fixed plans and $\rho = 0$, the difference between the original procedure and conservative speed rose from -3.1 points at $\sigma = 0.10$, where padding was slightly better, to 30.5 at $\sigma = 0.30$. Under strongly correlated delays the increase was not monotonic. The gain costs some resources: the original procedure used more distance than conservative speed (2.0–4.9%, depending on size), with fleet changes that went both ways (Section S10).

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Table 3
Paired differences in service probability (percentage points) on the same subsets as Table S3 of the SI. SC denotes screened conservative speed with its prespecified fallback.

n	S minus C	P minus C	SC minus C
20	−11.0 [−16.1, −6.0]	21.4 [9.6, 31.3]	28.2 [21.4, 34.3]
50	−13.1 [−17.4, −8.9]	17.9 [2.8, 32.0]	11.8 [2.7, 22.2]
100	−16.2 [−20.6, −11.7]	16.0 [4.8, 24.8]	11.2 [2.8, 19.4]
200	−12.9 [−18.5, −5.7]	8.3 [−5.1, 20.0]	5.7 [0.0, 12.9]

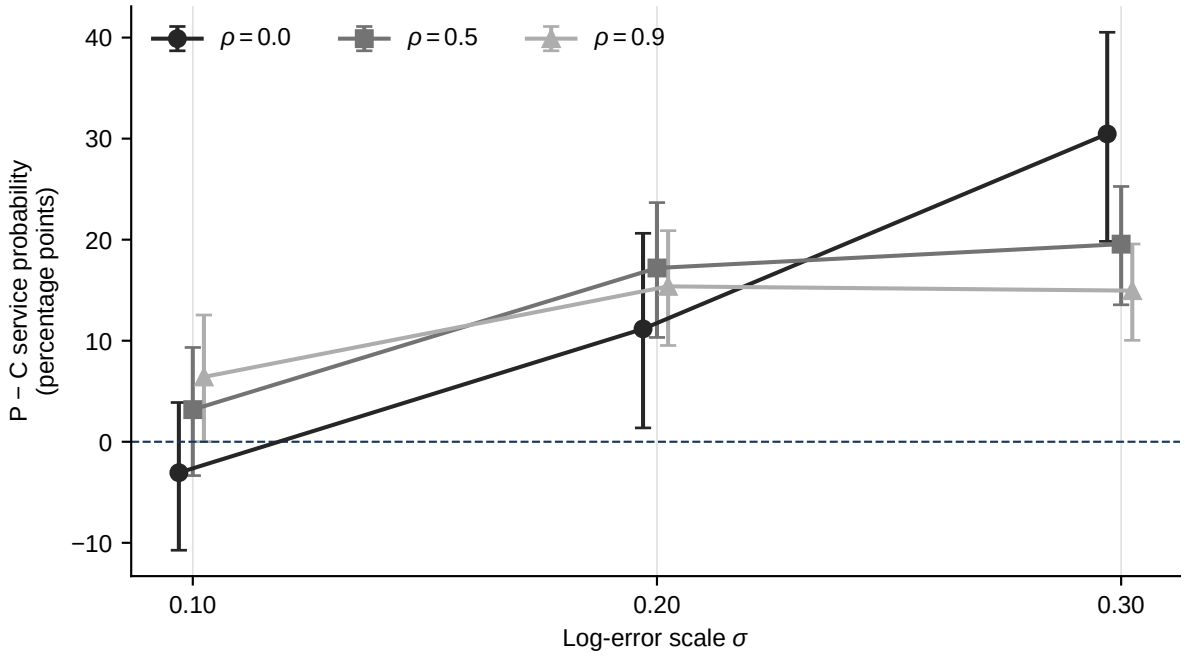


Fig. 1. Paired difference in service probability between the original procedure (P) and conservative speed (C) on the 46 common development instances, for fixed plans across the disturbance grid. Circles, squares and triangles mark $\rho = 0, 0.5$ and 0.9; error bars are 95% bootstrap intervals over instances, not simultaneous across cells.

5.2. Capping the contraction

Of the 21 development instances on which the original procedure admitted no plan, 17 met a certified reachability obstruction at some buffer level. Capping (Table 4) raised the number of admitted instances from 32 to 42 instances and pooled service by 7.7 [3.5, 12.5] points. At 100 and 200 customers the capped procedure was admitted on 18/18 instances and pooled 97.5–97.6%. It used 2.5% more distance and 0.43 more vehicles on average than the original procedure, rising to 1.88 vehicles at 200 customers.

The remaining shortfall comes from the fallback. Admitted plans averaged 97.4% validated service for the capped procedure, while its 11 fallback plans, all at 20 or 50 customers, averaged 45.2%. Of the admitted plans, 40 of 42 kept a lower bound of at least 0.95 on the fresh validation bank.

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Table 4
Development study, all 53 instances. For each procedure: instances admitted by screening out of all instances of that size, with the number whose lower bound also reached 0.95 on the validation bank in parentheses, and pooled validated service probability (%) with a 95% bootstrap interval, fallback plans included.

	Original procedure		Screened conservative		Capped procedure	
n	Admitted	Pooled	Admitted	Pooled	Admitted	Pooled
20	14/20 (14)	81.6 [70.2, 91.9]	15/20 (15)	87.7 [79.0, 94.7]	14/20 (14)	81.6 [70.3, 92.0]
50	8/15 (8)	72.7 [59.2, 86.4]	4/15 (4)	67.5 [58.2, 77.3]	10/15 (10)	80.1 [66.9, 90.8]
100	6/10 (5)	81.3 [68.8, 93.3]	4/10 (4)	77.5 [67.2, 88.2]	10/10 (9)	97.5 [96.7, 98.2]
200	4/8 (3)	80.6 [68.7, 92.3]	2/8 (2)	78.3 [71.0, 86.5]	8/8 (7)	97.6 [96.9, 98.2]

5.3. Confirmatory evidence

Table 5 lists every prespecified hypothesis. In protocol 1 the original procedure exceeded conservative speed by 15.7 points and capping added 11.7. Capping admitted 46 instances against 32 and lost none. H2 and H3 are therefore not independent: 11.6 of the 11.7 points of H2 come from the instances that only the capped procedure admitted. The capped procedure pooled 91.0% over all instances, against 79.4% for the original procedure.

Protocol 2 asked whether the gain survives a stronger generator. With HGS, screening with capped contraction exceeded HGS conservative-speed planning by 28.0 points, and pooled service reached 91.6% against 58.4%. The HGS and OR-Tools capped procedures differed by -0.2 points, an interval that bounds the difference but, with no equivalence margin set, does not show equivalence. Capping mattered more with HGS: at 100 and 200 customers it raised admission from 3 to 18 of 18 instances. The conservative fallback, suggested post hoc by protocol 1, did not measurably improve the capped OR-Tools procedure (G2): that procedure fell back on only 8 instances, and the new rule changed the plan on 4 of them, with effects in both directions. It did help the uncapped procedure (G3).

5.4. Comparison with robust optimization

On the 45 development instances with up to 100 customers, the standard robust model under box uncertainty had no plan on 16 instances at $\theta = 0.6$, each time because some customer could not be protected even on a direct trip; every smaller budget failed on the same instances. With relief, every setting had a plan, and the relieved settings traded distance for service along a smooth frontier (Fig. 2). Screened robust with relief pooled 97.2%, against 84.6% for the OR-Tools and 93.6% for the HGS capped procedure, and used 0.53 more vehicles than the latter.

Part A of protocol 3 confirmed this on the confirmatory instances with up to 100 customers. Screened robust with relief exceeded the OR-Tools capped procedure by 6.7 points over all 90 of them, and the HGS capped procedure by 6.7 points over the 45 of protocol 2, where HGS plans exist; the two estimates agree only by coincidence. It pooled 97.2% and used 0.42 more vehicles than the HGS procedure. Within the robust family, screening returned plans 1.3% shorter than the most protective setting for 1.2 points less service. When the robust plans were restricted, after the fact, to no more vehicles than the comparator's plan, they were not measurably more reliable: the differences were 3.2 [-1.1, 8.1] points

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Table 5
Every prespecified hypothesis of the three protocols. Primary hypotheses are marked *. Differences are in percentage points of validated service-event probability unless stated; intervals are paired bootstrap intervals over instances (20,000 resamples), at 97.5% for the two primary hypotheses of each study (Bonferroni) and 95% otherwise. A hypothesis is confirmed when it meets the decision rule fixed in its protocol.

Hyp.	Contrast	Estimate [interval]	Level	Decision
Protocol 1: 53 new instances, OR-Tools generator				
H1*	Original procedure minus conservative speed (50 instances)	15.7 [7.5, 23.6]	97.5%	confirmed
H2*	Capped minus original procedure (53 instances)	11.7 [5.9, 17.9]	97.5%	confirmed
H3	Instances admitted, capped versus original procedure	46 vs 32 ($p = 0.00012$)	exact	confirmed
H4	Advantage over conservative speed at $\sigma = 0.30$ minus at $\sigma = 0.10$, $\rho = 0$	28.3 [19.6, 36.4]	95%	confirmed
Protocol 2: 53 new instances, HGS and OR-Tools generators				
G1*	HGS capped procedure minus HGS conservative speed (49 instances)	28.0 [23.0, 32.8]	97.5%	confirmed
G2*	Conservative minus slack-aware fallback, OR-Tools capped procedure	0.4 [−0.7, 2.1]	97.5%	not confirmed
G3	The same for the uncapped OR-Tools procedure	3.8 [1.6, 6.2]	95%	confirmed
G4	HGS minus OR-Tools capped procedure, both with conservative fallback	−0.2 [−4.0, 3.7]	95%	estimate
Protocol 3, part A: 90 confirmatory instances with up to 100 customers				
R1*	Screened robust with relief minus OR-Tools capped procedure	6.7 [3.0, 11.1]	97.5%	confirmed
R2*	Screened robust with relief minus HGS capped procedure (45 instances)	6.7 [2.2, 12.2]	97.5%	confirmed
R3	The same contrast in vehicles	0.42 [0.20, 0.67]	95%	confirmed
R4	Distance (%) of screened robust versus the most protective relieved setting	−1.3 [−1.7, −0.9]	95%	confirmed
Protocol 3, part B: 53 new instances, best-screened fallback				
F1*	Best-screened minus slack-aware fallback, OR-Tools capped procedure	7.3 [2.3, 13.2]	97.5%	confirmed
F2*	Best-screened minus conservative fallback, HGS capped procedure	4.2 [1.2, 8.0]	97.5%	confirmed
F3	Pooled service (%), OR-Tools capped procedure with the new rule	94.4 [90.1, 97.1]	95%	estimate
F4	Pooled service (%), HGS capped procedure with the new rule	94.9 [91.0, 97.1]	95%	estimate

against the HGS and −0.7 [−3.9, 2.7] against the OR-Tools capped procedure. On these instances the screened procedures trade some reliability for fewer vehicles, returning the shortest candidate that passes a stated test. Runs with 200 customers are reported in Section S10.12.

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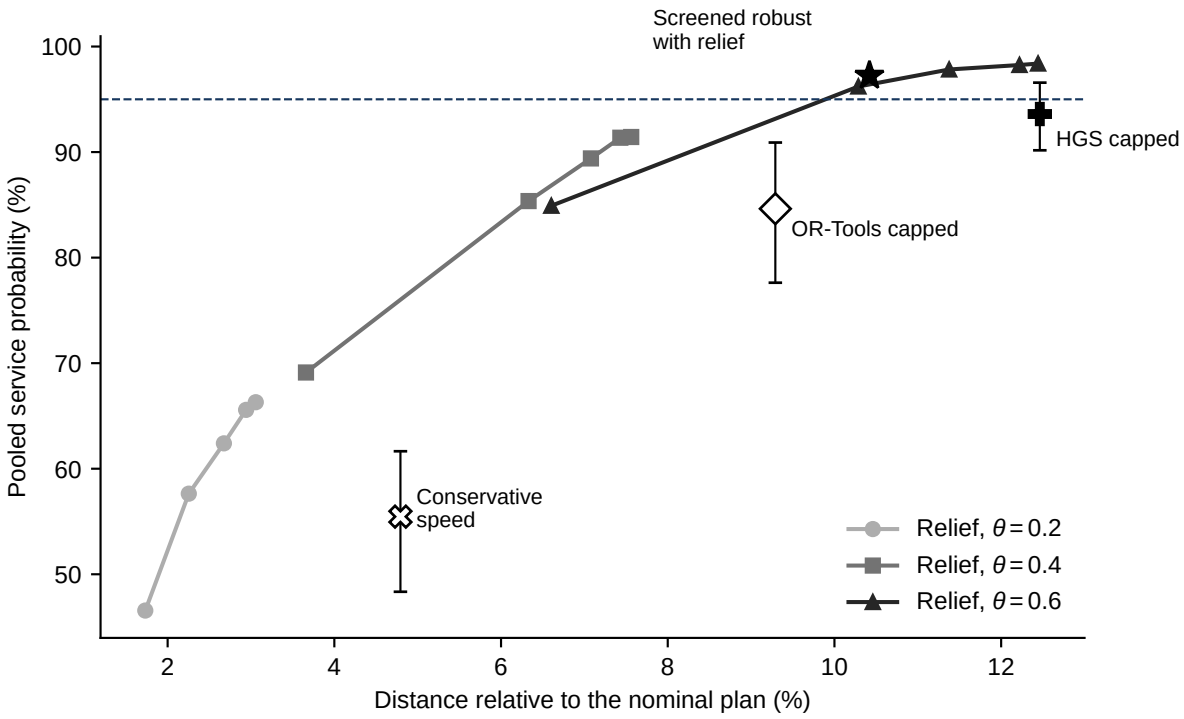


Fig. 2. Pooled validated service probability against mean distance relative to the nominal robust plan, on the 45 development instances with up to 100 customers. Lines join the relieved robust settings for each θ, from $\Gamma = 1$ to box uncertainty; the screened procedures and conservative speed are shown with 95% bootstrap intervals. The dashed line marks the target.

5.5. Fallback rule

After the first two protocols, the stored records showed that returning the best-screened candidate when nothing is admitted raised pooled service in every study and procedure, by 3.0 to 15.6 points. Part B of protocol 3 tested the rule on 53 new instances. Over each procedure's default fallback, it improved the OR-Tools capped procedure by 7.3 points and the HGS capped procedure by 4.2, to pooled service of 94.4% and 94.9%. Both estimates stay just below the 95% target. On 1 instance the anchor solves found no plan, so neither procedure could run and it scored zero; without it, pooled service was 96.3% and 96.7% (descriptive). Plans returned by the fallback have not passed the screen.

5.6. Sensitivity and further checks

Objective weights. On the development instances, equal weights admitted fewer instances (23 against 31 in the same run) and cut the advantage over conservative speed to 6.3 points [−1.1, 13.7]. A later run on another machine, repeating both vectors, cut it from 17.5 to 8.1 points [1.1, 15.1]. Raising the

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driver-duration weight alone changed service by -8.9 points and raising the fleet weight alone by -5.6, so the advantage depends on how much waiting time the objective leaves to absorb delay. No protocol varied the weights (Section S10.8).

Benchmark instances. On 8 Solomon R1 and RC1 instances, OR-Tools found a plan for 5 and screening admitted 3. With HGS on 20 Gehring–Homberger instances with 200 customers (Gehring and Homberger, 1999), the procedure could run on 18, admitted 17, and exceeded conservative speed by 15.1 points [11.0, 19.1] (Sections S10.9 and S10.11).

Delay-model misspecification. Under four alternative delay models, the fixed plans of the capped procedures lost up to 6.9 points and their advantage over conservative speed held. Screening the stored candidates again under each model, with the best-screened fallback, kept pooled service between 93.2 and 96.7% (Section S14.3).

Family-wise screening and computation time. Over all studies and procedures, a Bonferroni bound over the 13 buffer levels removed all 11 of the 318 admitted plans that later missed the validation bound, while 1 newly selected plan missed it, at the cost of at most 2 admissions per study and procedure and between -1.8 and 0.2 points of pooled service. Screening a candidate took 3 to 24 ms, against seconds to generate it (Section S14.1).

6. Discussion and limitations

For a planner who already uses a VRPTW solver, the results suggest a practical change. Generating a handful of buffered variants and screening them on simulated days costs little beyond the solves, needs no change to the solver or its objective, and replaced a fixed 20% pad with plans that were measurably more reliable on the same instances. Each admitted plan has a tested lower bound on a defined service event, and the validation bank reports how that plan behaved on days it had not seen. The reachability cap is a one-line change that removed most certified failures and should be used with any buffer-based method, robust models included.

The evidence has clear limits. The instances are synthetic and the disturbance parameters are assumed; they have not been calibrated to vehicle GPS data, so the numbers are not forecasts for a real fleet. The protocols test repeatability within one generator and one delay model, and the Solomon and Gehring–Homberger checks are small. The OR-Tools generator is weaker than the state of the art, which is why protocol 2 repeated the test with HGS; two generators still do not show that the result holds for every solver. The robust comparison covers one budgeted model with one search configuration, and the distributionally robust and riskiness-index methods (Jaillet et al., 2016; Zhang et al., 2019) are not compared. The advantage depends on the objective weights, and equal weights more than halved it. The Wilson bound is a per-candidate guarantee, not a simultaneous one over the grid. Finally, the evaluation keeps every departure fixed and excludes breakdowns, new orders and rolling-horizon replanning, so it is stricter than an operator who adjusts departures would face, and it says nothing about recovery once a delay occurs.

Under the prespecified fallback rules, pooled service stayed below 95% in every study because of instances on which nothing was admitted. The best-screened fallback narrows the gap, but closing it needs better candidates on those instances, which is where a stronger or stochastic generator could add most.

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7. Conclusion

Screening a family of buffered plans from a standard VRPTW solver gave more reliable service than uniform travel-time padding on matched synthetic instances, with OR-Tools and with HGS, and three timestamped protocols supported the main findings. One primary hypothesis was not confirmed: returning the conservative-speed plan when nothing is admitted did not measurably help the capped OR-Tools procedure, while returning the best-screened candidate did. Capping each customer's buffer at its direct-reachability limit removed a failure that also stops robust models from finding any plan. A budgeted-robust optimizer with that relief was more reliable still, but used more vehicles, and its advantage was not measurable at the comparator's fleet size. The screening layer and the reachability relief worked with every planner tested. Calibrated delay data and real road networks are the next test.

Data availability

All code, generated instances, raw records and analysis scripts are openly available on Zenodo (Ali, 2026d); code is released under the MIT License and records under CC BY 4.0. The three protocol freezes are separate Zenodo records (Ali, 2026a,b,c). The Solomon and Gehring–Homberger instance files are third-party data (Solomon, 1987; ML4VRP, 2024; Gehring and Homberger, 1999) and remain under their original terms.

Use of generative AI

Generative AI assistants (OpenAI's ChatGPT and Anthropic's Claude) were used for code development, code review, checking the analysis, and drafting and editing text. They are not authors. Every experiment was run from the archived code and every reported number regenerates from the archived records. The author reviewed all code and text and takes full responsibility for the content.

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